Shared Neural Response Function and Activation Estimate via Bayesian Modeling

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From last meeting

- Learn basics of intersubject correlation analysis
- Find simultaneous EEG/fMRI neuroimage data involving naturalistic stimuli
- Consider methods for multi-subject and multi-group studies
- Develop methodology for detecting difference between two groups

Shared Neural Response Theory

At a given voxel, the fMRI signal[1] $y_i(t)$ for subject i at time t is

$$y_i(t) = \alpha_i c(t) + \beta_i id(t) + s_i \tag{1}$$

where components include

- $c(t)$, consistent across all subjects
- $id_i(t)$, idiosyncratic response of subject i
- s_i spontaneous noise

[1] Nastase et al. 2019

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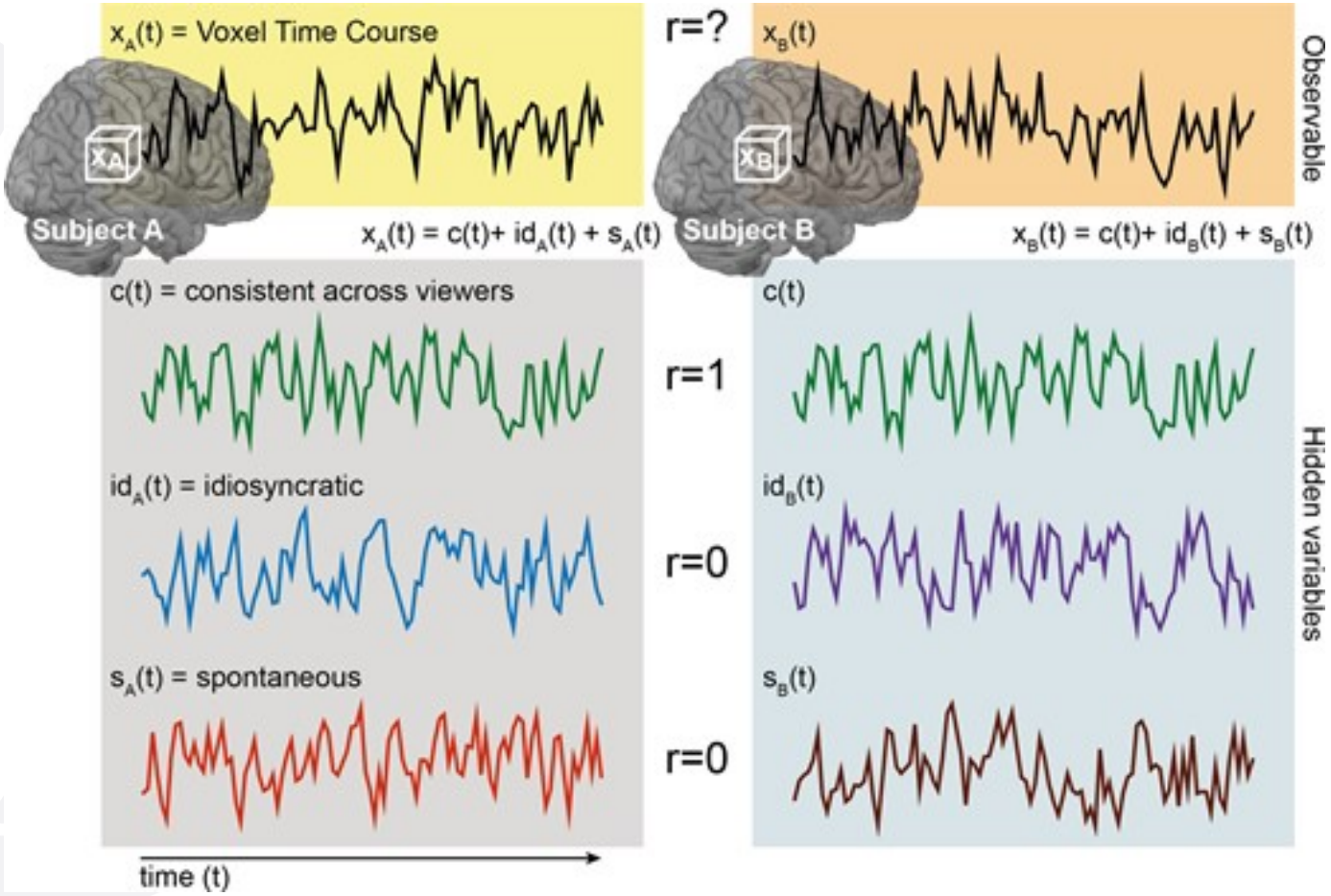
Current



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Basic Intersubject correlation analysis

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At each voxel

i. Calculate pairwise Pearson correlation between every subject
ii. Transform correlations and calculate mean/median
iii. Test for activity via nonparametric resampling methods

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iii. Test for activity via nonparametric resampling methods

or

i. Leave one subject out
ii. Calculate mean signal for remaining subjects
iii. Measure distance/correlation between left out signal and remaining mean

Drawbacks

- Serious multiple testing issues, even using resampling methods
  - Narrow sampling distributions
  - High false positive rate
- Large information loss in using summary statistics
- No borrowing information from neighboring voxels
- Lack of uncertainty quantification in function estimates
- No statistical model

Proposed Alternative

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Fully Bayesian model based analysis of naturalistic neural response

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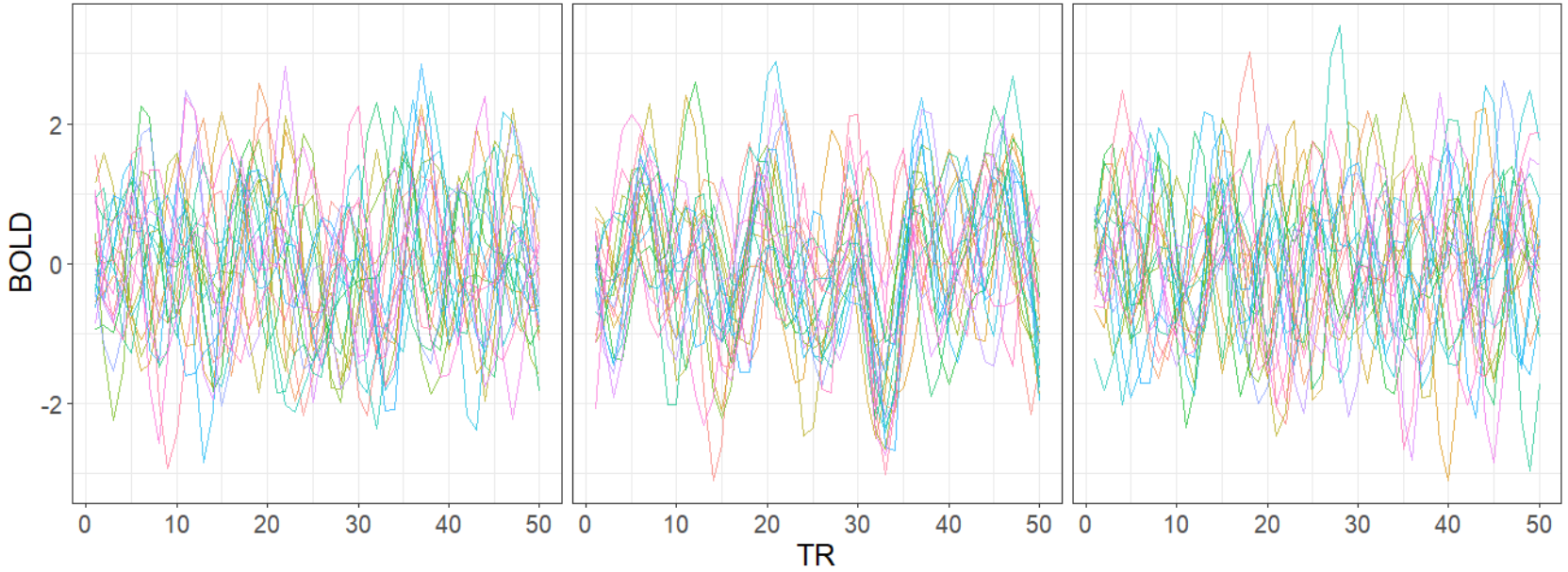
Fully Bayesian model based analysis of naturalistic neural response

$$y_{v,i}(t) = \beta_v f(t) + \epsilon_{v,i}(t)$$

- β_v is constructed using a spatial prior which also induces sparsity (e.g. only large/significant if there is a clearly shared signal at voxel v). Shared by all subjects
- $f(\cdot)$ is estimate of $c(\cdot)$ shared over whole brain/ROI (using Gaussian process prior)
- $\epsilon_{v,i}(t)$ captures idiosyncratic response and spontaneous noise

Single Voxel Signals



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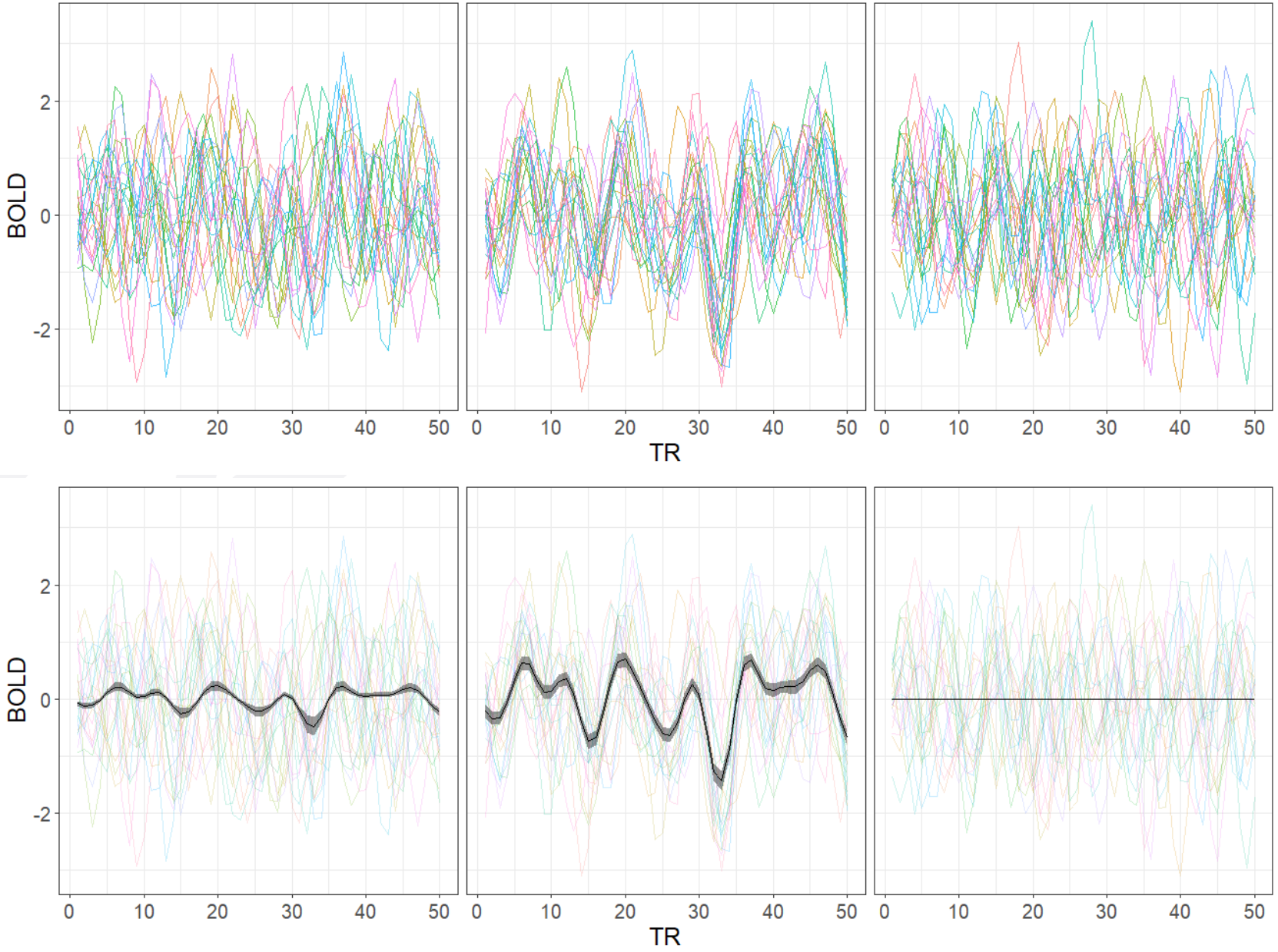
Single Voxel Signals



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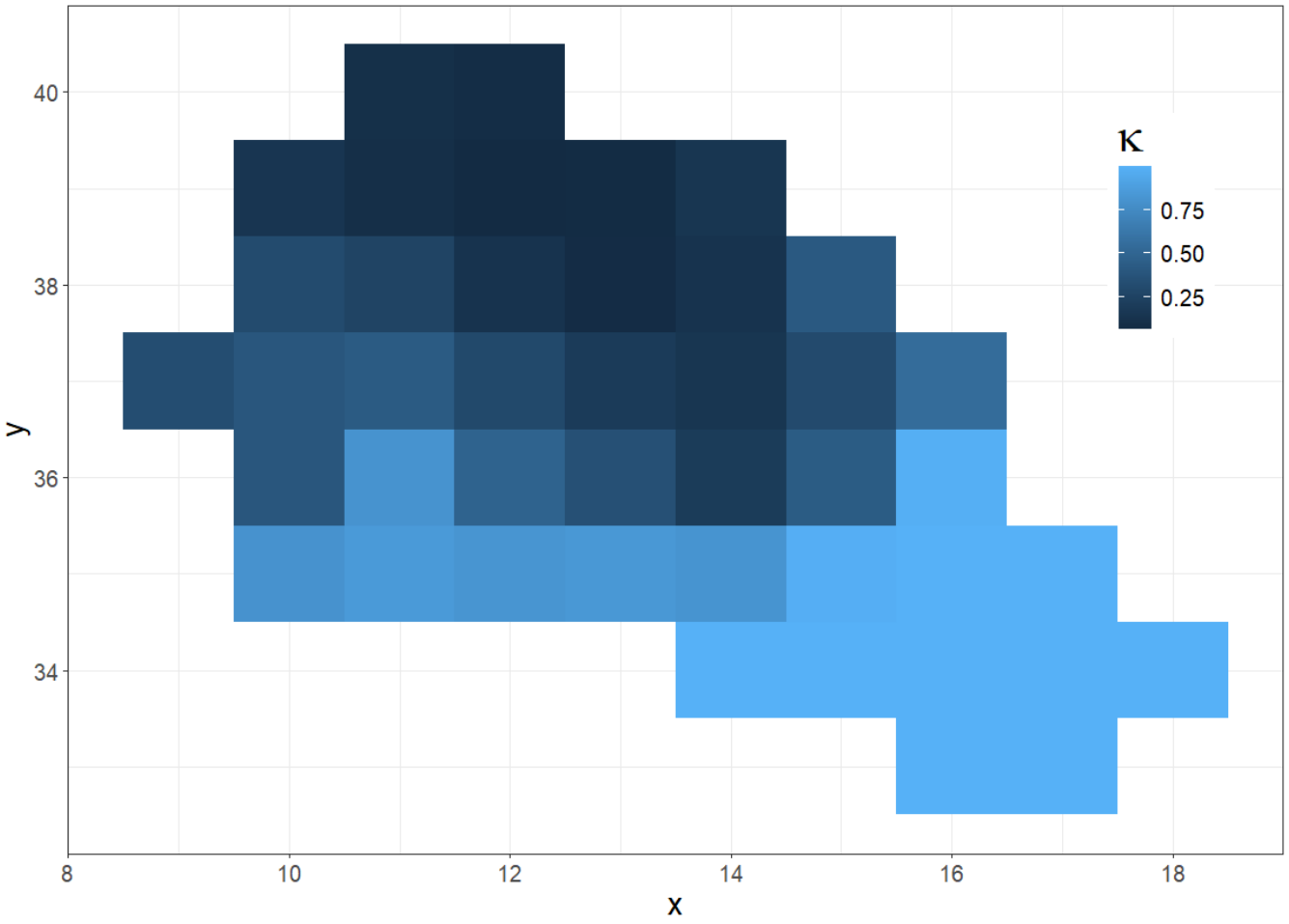
Activated Voxels



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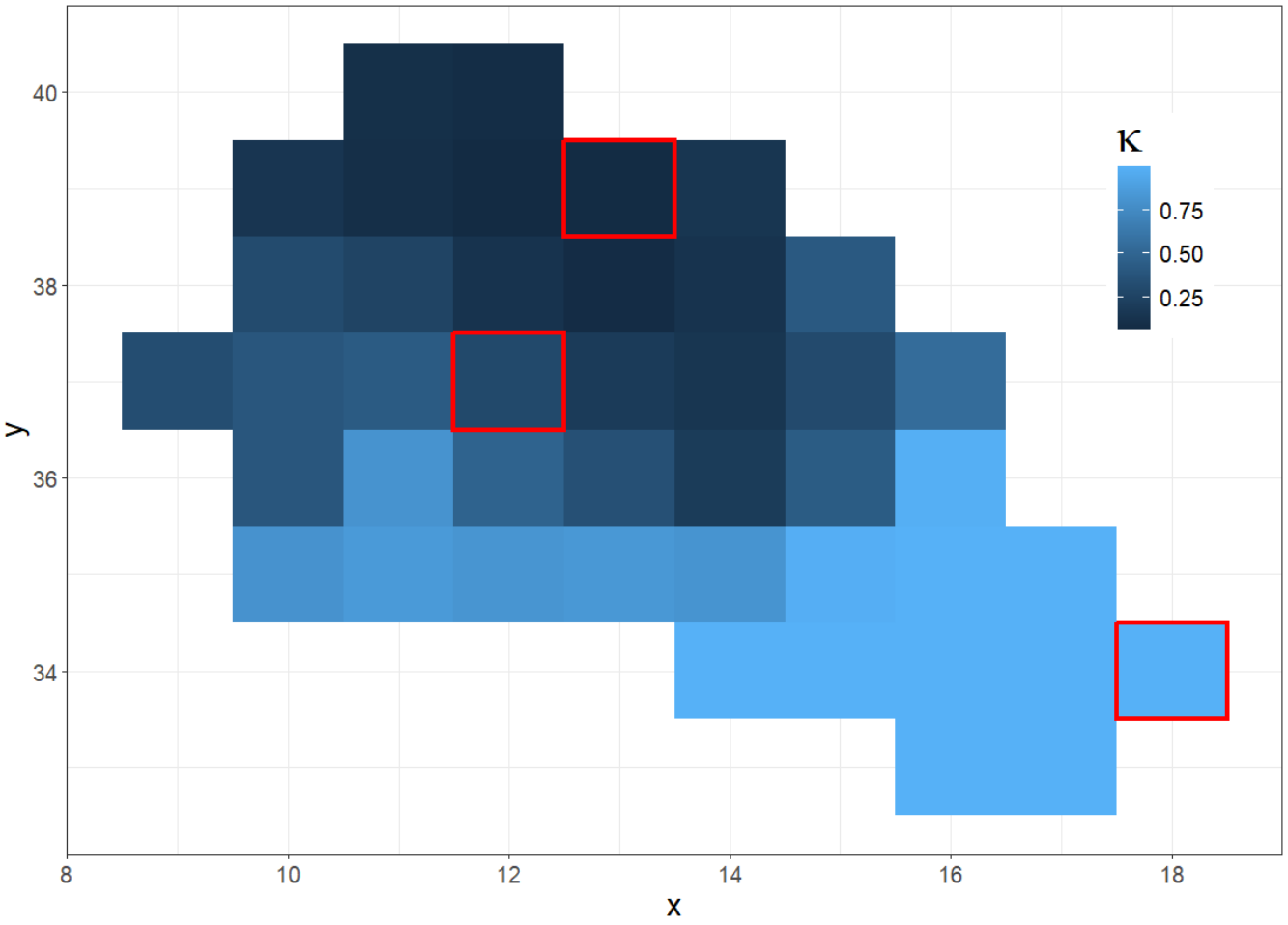
Activated Voxels



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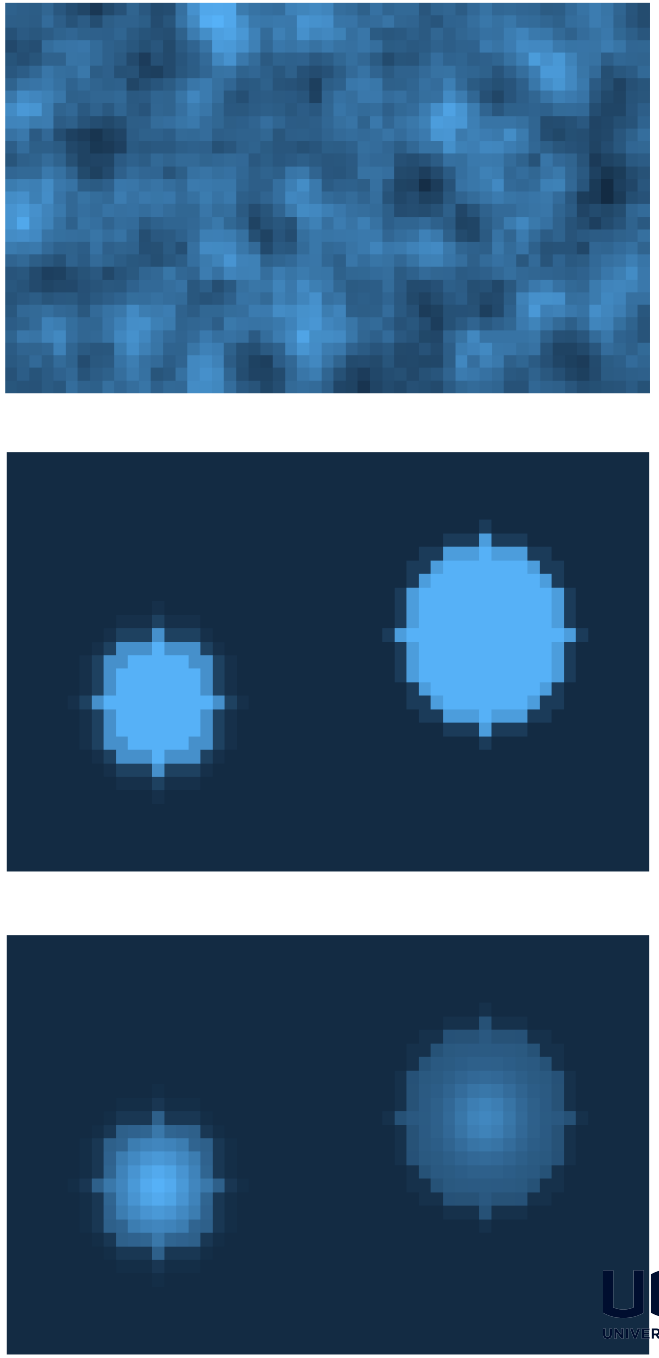
Simulation Data

Data

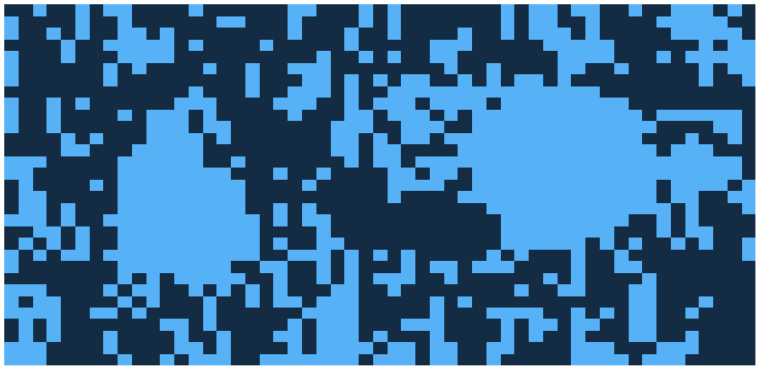
- 22 subjects
- Shared grid of 53 $\times$31" *voxels*"

- Activated voxels and signal strength known
- Simulate BOLD fMRI for 200 seconds (TR = 2)

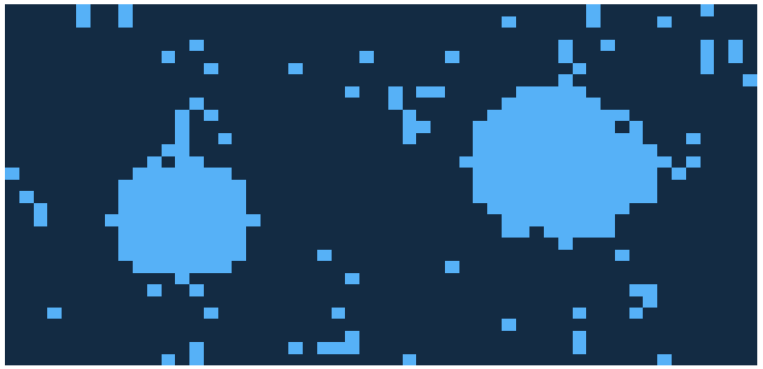


Preliminary Simulation Results

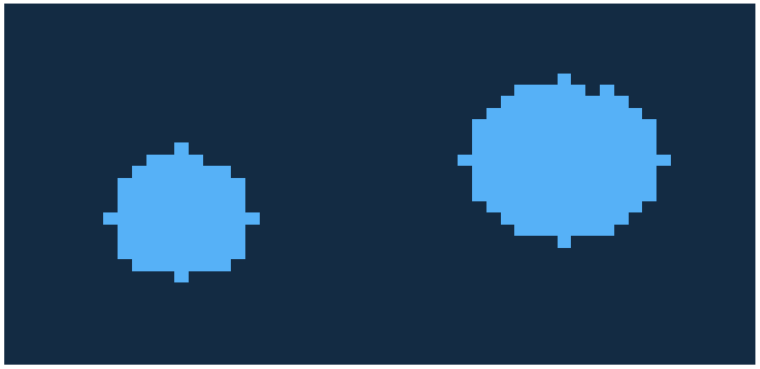
Cyclic



Scramble



Model



fMRI Data

Data from Telesford et al. 2023

- 22 adult subjects
- 105 seconds of Despicable Me
- Scans registered
- Voxels separated into ROIs

fMRI Data

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Real Data Results

Cyclic

Scramble

Model

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To Do

- Clean/formalize statistical model
- Full scale simulation study
- Extend to two group analysis (e.g. children with autism vs. typically developing children)
  - Independent model for each group
  - Bayesian multigroup/hierarchical Gaussian process[2]

[2] Li et al. 2025

References

- Nastase, Samuel A., et al. "Measuring shared responses across subjects using intersubject correlation." Social cognitive and affective neuroscience 14.6 (2019): 667-685.
- Carvalho, Carlos M., Nicholas G. Polson, and James G. Scott. "The horseshoe estimator for sparse signals." Biometrika (2010): 465-480.
- Datta, Abhirup, et al. "Hierarchical nearest-neighbor Gaussian process models for large geostatistical datasets." Journal of the American Statistical Association 111.514 (2016): 800-812.
- Telesford, Qawi K., et al. "An open-access dataset of naturalistic viewing using simultaneous EEG-fMRI." Scientific Data 10.1 (2023): 554.
- Li, Didong, et al. "Bayesian Multi-Group Gaussian Process Models for Heterogeneous Group-Structured Data." Journal of Machine Learning Research 26.30 (2025): 1-34.